Online Approval Committee Elections

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Agenda

New restaurant in town

Dim Sum

- 3 friends : Alice, Bob and Charlie
- No menu, dishes to share
- Waiter shows one at the time
- Friends approve/ disapprove dishes collectively decide

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- Online selection
- Final menu = Committee
- Decision as a group → Approval Election

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Some possible instance

2 dishes to select (budget)

- Cheese (0,1,1)
- Risotto (1,1,0)
- Sushi (1,0,0)
- Cake (1,1,1)
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Maximizing Aggregated Satisfaction

- 3 Thiele methods : AV, αCC, PAV
- How to evaluate the contribution of future candidates ?
- Number of remaining seats and candidates, committee, score
- Using MDPs to find best policies

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- Can we ensure some axiomatic properties ?
- 3 properties studied JR, PJR, EJR

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- Impossibility of EJR
- Introducing α-EJR
- Tight bound for approximating EJR

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